

Operating System

- An operating system is a set of basic system programs that manage computer hardware to perform basic computer functions, that is, it enables a connection between hardware and user programs.
- It is a type of software that interface between ***user and computer hardware devices***.
- It performs all the basic tasks like file management, memory management, process management, handling input and output, and controlling peripheral devices such as disk drives and printers.
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- Some popular Operating Systems include **Linux, Windows , macOS, iOS, Android**, etc.

Linux

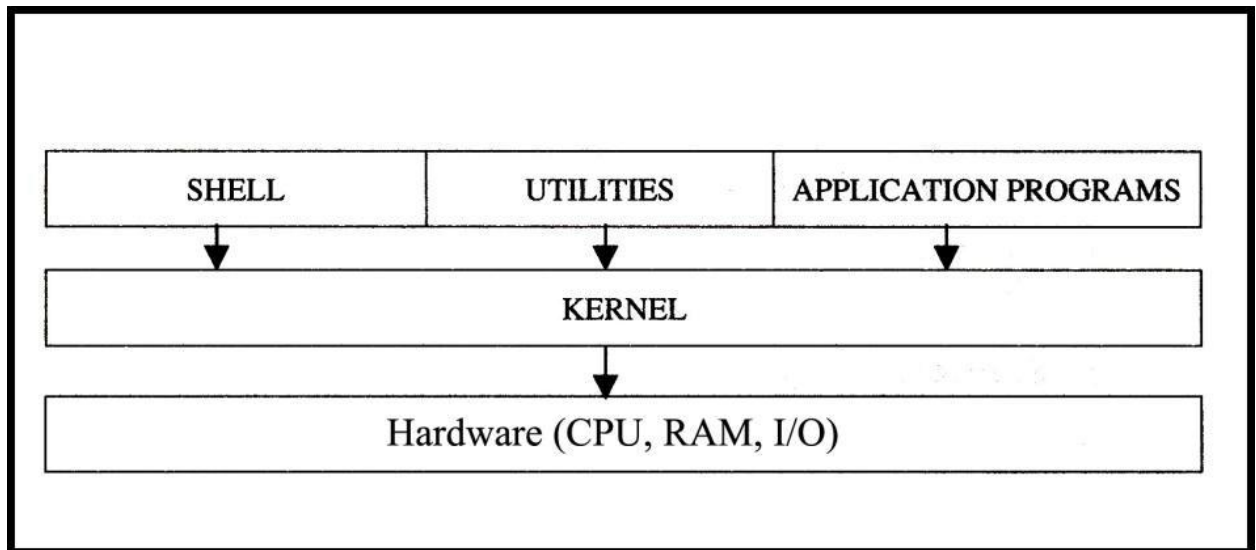
- Linux is a free, open-source operating system kernel that serves as the foundation for a variety of operating systems known as Linux distributions.
- It was originally developed by Linus Torvalds in 1991 and has since become one of the most widely used operating systems, especially in servers, embedded systems, and increasingly in desktop environments.

- It is freely available for commercial as well as for non-commercial purposes. As it is available for commercial purposes that means any programmer can do some changes in the Linux kernel by doing some code and finally make it as the new distribution.
- *Distributions (Distros):*
- Linux distributions bundle the Linux kernel with a package management system, software, and utilities.

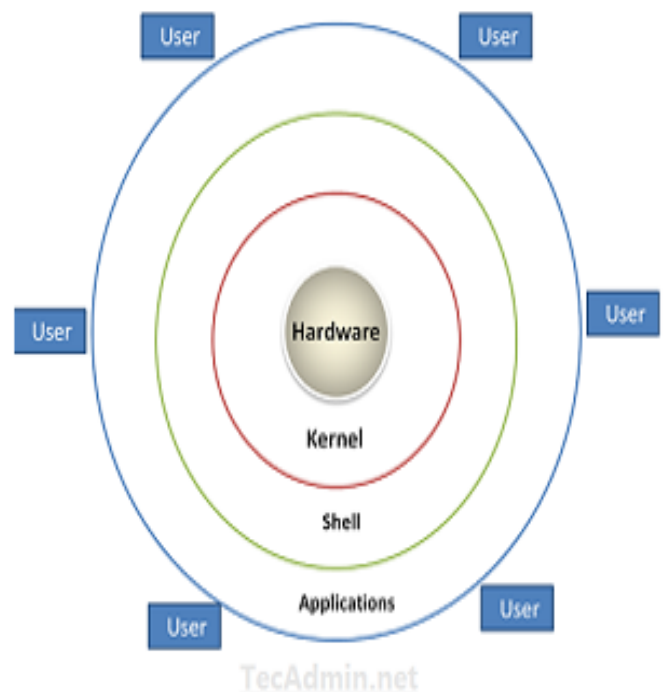
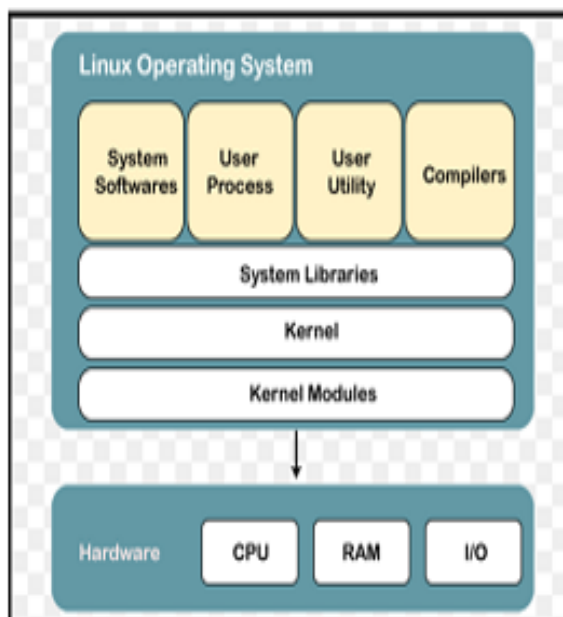
Popular Linux distros include:

- Ubuntu: A user-friendly distribution based on Debian, popular for desktop use.
- Debian: A stable, non-commercial distribution focused on free software.
- Fedora: Sponsored by Red Hat, known for using the latest software versions.
- CentOS: A free alternative to Red Hat Enterprise Linux (RHEL).
- Arch Linux: Known for its simplicity and customizability.
- Mint: Based on Ubuntu, offers a polished, user-friendly experience.

- **Linux Architecture**



- The Linux operating system's architecture contains some of the components: **the Kernel, System Library, Hardware layer, System, and Shell utility.**



1. Kernel:

- **The core** of the Linux operating system, responsible for managing hardware resources like CPU, memory, and I/O devices.
- It provides a bridge between applications and the actual hardware.
- The Linux kernel is monolithic, meaning all the core functionality is part of one large block of code, but it can dynamically load modules.
 - Monolithic Kernel- A monolithic kernel is a type of kernel where the entire operating system runs in kernel space (privileged mode). All the core services, such as process management, memory management, file systems, and device drivers, run in a single address space (the kernel space).
 - Micro kernels- Microkernels aim to minimize the kernel's size by only including the most essential services in kernel space. These essential services usually include process management, memory management, and inter-process communication (IPC). Other services, such as device drivers, file systems, and network stacks, run in user space.
 - Exo kernels- An exokernel is a minimalistic kernel architecture that aims to provide direct access to the hardware without imposing abstractions typically

seen in traditional operating systems. Instead of providing high-level abstractions (such as file systems or network protocols), exokernels expose raw hardware resources directly to applications, allowing them to define their own abstractions.

- Hybrid kernels- A hybrid kernel attempts to combine the benefits of both monolithic and microkernel architectures.

Kernel Type	Description	Example Systems	Pros	Cons
Monolithic Kernel	All services (drivers, memory, filesystems) run in kernel space.	Linux, Unix	High performance, direct communication between components	Less modular, a crash can bring down the entire system.
Microkernel	Only essential services run in kernel space, others (like drivers) run in user space.	Minix, QNX	Stable, easy to maintain, service isolation	Slower due to context switches and IPC overhead.
Exokernel	Provides no abstractions, giving applications direct access to hardware resources.	MIT Exokernel	High performance, flexibility in defining abstractions	Complex, harder to develop applications.
Hybrid Kernel	Mixes monolithic and microkernel designs, with critical services in kernel space and others in user space.	Windows NT, macOS (XNU Kernel) ↓	Balance of performance and modularity	Increased complexity.

- **2. System Libraries:-** System libraries are collections of reusable code that programs and applications in an operating system can use to perform common tasks. In the

context of Linux and Unix-like systems, system libraries provide essential functionality such as interacting with the operating system kernel, handling input/output, performing mathematical operations, managing files, and more.

- **3. System and shell Utility :-** System utilities in Linux are tools and programs that allow users and administrators to interact with the operating system, manage resources, monitor performance, and perform various administrative tasks. These utilities are essential for maintaining and configuring the system, managing files, monitoring processes, and ensuring system security and performance.

1. File and Directory Management Utilities:

These utilities help users manage files, directories, and permissions.

- **ls:** Lists files and directories.
- **cp:** Copies files and directories.
- **mv:** Moves or renames files and directories.
- **rm:** Removes files and directories.
- **mkdir:** Creates a new directory.
- **rmdir:** Removes empty directories.
- **chmod:** Changes file permissions.
- **chown:** Changes the ownership of files or directories.
- **find:** Searches for files and directories based on various criteria.
- **df:** Displays disk space usage.
- **du:** Shows the size of files and directories.

2. Process Management Utilities:

These utilities help manage system processes, including monitoring, starting, and stopping processes.

- **ps**: Displays information about active processes.
- **top** / **htop**: Displays real-time information about system processes and resource usage.
- **kill**: Sends signals to processes, often used to terminate processes.
- **killall**: Kills processes by name.
- **bg** / **fg**: Moves jobs to the background (bg) or foreground (fg).
- **nice** / **renice**: Sets or changes the priority of a process.
- **systemctl**: Controls system services and daemons (used with systemd).

3. Disk and Filesystem Utilities:

Utilities that help manage and maintain disks, partitions, and filesystems.

- **fdisk**: Command-line utility for partitioning hard drives.
- **mkfs**: Creates a new filesystem on a partition (e.g., mkfs.ext4).
- **fsck**: Checks and repairs filesystem inconsistencies.
- **mount**: Mounts a filesystem or device.
- **umount**: Unmounts a filesystem or device.
- **blkid**: Displays block device attributes, such as UUIDs.
- **tune2fs**: Tunes or modifies the settings of an ext2/ext3/ext4 filesystem.
- **parted**: A disk partitioning tool.

4. User Management Utilities:

These utilities help manage users, groups, and permissions.

- **useradd**: Adds a new user to the system.
- **usermod**: Modifies an existing user's account.
- **userdel**: Deletes a user account.
- **passwd**: Changes a user's password.
- **groupadd**: Adds a new group.
- **groupdel**: Deletes a group.
- **id**: Displays user ID and group information.
- **whoami**: Displays the current username.

5. System Monitoring Utilities:

These tools provide insights into system performance, including CPU, memory, network usage, and overall system health.

- **uptime**: Shows how long the system has been running and system load averages.
- **free**: Displays information about memory usage.
- **vmstat**: Displays virtual memory statistics and performance metrics.
- **iostat**: Reports CPU and I/O statistics.
- **sar**: Collects and reports system activity and performance statistics.
- **netstat**: Displays network connections, routing tables, and interface statistics.
- **dstat**: Provides comprehensive system resource statistics.
- **iostat**: Reports CPU and input/output statistics for devices.
- **mpstat**: Reports individual CPU core statistics.

6. Networking Utilities:

These utilities help configure and monitor the network, test connectivity, and troubleshoot network-related issues.

- **ifconfig**: Configures network interfaces (deprecated in favor of ip command).
- **ip**: A more modern tool to configure network interfaces, routing, and more.
- **ping**: Tests network connectivity between systems.
- **traceroute**: Traces the route packets take to a network destination.
- **netstat**: Displays active network connections and listening ports.
- **ss**: Displays socket statistics, a more modern alternative to netstat.
- **dig**: DNS lookup tool to query DNS records.
- **nslookup**: Another DNS lookup tool.
- **tcpdump**: Captures and analyzes network packets.
- **iptables** / **nftables**: Configures the firewall rules to control network traffic.

7. System Boot and Service Management:

Tools that help manage system startup processes, services, and daemons.

- **systemctl**: Manages systemd services, used to start, stop, and manage system services.
- **service**: Manages services on older systems that don't use systemd.

- **journalctl**: Views logs generated by systemd and its services.
- **cron**: A daemon used to schedule recurring tasks.
- **at**: Schedules one-time tasks to be run at a specified time.
- **init**: Manages the system initialization process (replaced by systemd in many Linux distros).
- **grub**: Bootloader used to manage multi-boot systems and configure the kernel.

8. Package Management Utilities:

Package management tools help install, update, remove, and manage software packages on the system.

- **apt-get / apt**: Command-line package management tool for Debian-based distributions (e.g., Ubuntu).
- **yum / dnf**: Command-line package management tools for Red Hat-based distributions (e.g., CentOS, Fedora).
- **rpm**: A package manager for Red Hat-based systems.
- **pacman**: The package manager for Arch Linux and its derivatives.
- **snap**: A package management system for Linux that enables installation of software packages in a universal format.
- **flatpak**: Another system for managing universal Linux software packages.

9. Backup and Compression Utilities:

Utilities for archiving, compressing, and backing up files and directories.

- **tar**: Combines multiple files into one archive file (often with compression).
- **gzip** / **gunzip**: Compresses and decompresses files using the gzip format.
- **bzip2** / **bunzip2**: Compresses and decompresses files using the bzip2 format.
- **zip** / **unzip**: Compresses and decompresses files in the zip format.
- **rsync**: Synchronizes files and directories between two locations.
- **dd**: Used for low-level copying and backup of disks and partitions.

10. Text Processing Utilities:

These utilities help process and manipulate text files, often used in shell scripting and automation.

- **cat**: Concatenates and displays file contents.
- **grep**: Searches for patterns within text.
- **awk**: A powerful pattern scanning and processing language.
- **sed**: A stream editor for editing text in-place.
- **cut**: Extracts sections of each line in a file.
- **sort**: Sorts lines in text files.
- **wc**: Word count utility that counts lines, words, and characters in files.
- **head** / **tail**: Outputs the first/last parts of files.

11. System Information Utilities:

These utilities provide information about the system's hardware, software, and configuration.

- **uname:** Displays system information such as kernel name and version.
- **hostnamectl:** Displays and sets the hostname of the system.
- **lshw:** Lists detailed information about the hardware configuration.
- **lsblk:** Lists information about all available block devices.
- **lscpu:** Displays information about the CPU architecture.
- **dmidecode:** Extracts hardware information from the system BIOS.
- **lsusb:** Lists USB devices.
- **lspci:** Lists PCI devices.
- **dmesg:** Displays system messages, especially useful for debugging hardware issues.

4. Hardware layer:- This operating system contains a hardware layer that consists of several peripheral devices like CPU, HDD, and RAM.

Shell:-

The primary function of the shell is to execute user commands.

- Commands like file operations (ls, cp, mv), process management (ps, kill), and system information (df, free) are handled by the shell.
- It is an interface among the kernel and user.
- The shell interprets user input, translates it into instructions for the system, and provides the output or performs the desired task.
- It can take commands through the user and runs the functions of the kernel. The shell is available in distinct types of operating system's. These operating systems are categorized into two different types of shells, which are the **graphical shells** and **command-line shells**.
- The graphical line shells facilitate the graphical user interface, while the command line shells facilitate the command line interface. Thus, both of these shells implement operations. However, the graphical user interface shells work slower as compared to the command-line interface shells.

Common Linux shells include:

- **Bash** (Bourne Again Shell) is the default in many distros.
- **Zsh, Fish, Ksh**, and others are also available for advanced users.

Benefits of Linux

- It offers a free operating system on the internet.
- It is easy to learn Linux for beginners
- Being open-source, anyone with programming knowledge can modify it.
- Multitasking operating system: can run multiple job simultaneously without affecting the system speed
- Capable of handling large no. of users simultaneously
- **security** : It's not completely safe but less vulnerable than others, does not required any antivirus program
- Once you have Linux installed you no longer need an antivirus! Linux is a highly secure system. **Virus** is not executed until the administrator send the Password
- Light weight-less requirement for running Linux
- Linux freeware is the OS of choice for Server environments due to its stability and reliability (Mega-companies like

Amazon, Facebook, and Google use Linux for their Servers)..

- Provide high performance over a different networks
- Flexible, can be used for desktop/server app ,embedded system
- Software updates are in user control: more faster updates, system update continuously without failing any issues
- programmers can design their own custom operating system

Why the Linux is better than other OS?

- There are many features of Linux that demonstrate that it is better than other operating system
- ***Free to Use:-*** Linux is accessible to the public for free. However, that is not the case with Windows.
- ***Variety Of Distributions:-*** You will find tons of Linux distributions catered for a different set of needs.
Customization :- Apart from installing themes , you have tons of beautiful icon themes. You can also do a lot around wallpapers in Linux.
- ***Software update***

- ***Open Source Nature :-***
- ***Secure :-*** Linux does not require the use of such Anti-Virus programs. Being secure by nature would make things easier for you and save money as well because you will not have to get a premium Antivirus software.
- ***Reliability :-***
- ***perfect for programmers :-*** Linux supports almost all of the major programming languages (*Python, C/C++, Java, Perl, Ruby, etc.*).

Disadvantages:

- One of the main disadvantages of Linux is that it can have a steeper learning curve compared to other operating systems
- **Software compatibility:** One more potential disadvantage of Linux is that some of the software and applications may not be readily available.
- **Hardware Compatibility:** In Linux, certain hardware devices have limited or no support at all. This can be due to several reasons, such as the device not having an open-source driver or the manufacturer not providing Linux-compatible drivers.

- **Lack of Standardization:** Another point that falls under the disadvantages of Linux is the lack of standardization in the ecosystem.

Lab work

- Linux Directory Commands
 - Linux File Commands
 - Linux File Content Commands
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